James Preston

Senior Data Engineer – AI/ML

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SUMMARY

Senior Data Engineer with over 10 years of experience building and maintaining cloud-based data platforms in healthcare, fintech, and enterprise environments. Skilled in designing scalable ETL/ELT pipelines, real-time streaming architectures, and data warehouses using AWS, Azure, Snowflake, and Databricks. Experienced in supporting AI and ML workloads, including NLP, feature store pipelines, and clinical data applications. Strong background in data modeling, governance, security, and observability, with hands-on experience leading teams, mentoring engineers, and delivering reliable, production-ready data solutions.

SKILLS

- **Programming & Query Languages:** Python (pandas, PySpark, boto3, requests, hl7apy, NumPy), SQL (Snowflake, Redshift, Synapse, Postgres, Aurora), Spark SQL, Bash
- **Data Engineering & Distributed Processing:** Apache Spark (Batch & Structured Streaming), Databricks (Delta Lake), AWS Glue, Azure Data Factory, Hadoop MapReduce, Hive
- **Data Warehousing & Modeling:** Snowflake (RBAC, clustering, dynamic masking), Amazon Redshift, Azure Synapse Analytics, PostgreSQL, Aurora, Azure SQL Data Warehouse, Dimensional Modeling (Star Schema, Snowflake Schema), Hybrid Star + Data Vault, Slowly Changing Dimensions (Type 2), Bronze–Silver–Gold Architecture
- **Streaming & Event-Driven Systems:** Apache Kafka, Kafka Connect, AWS Kinesis, Azure Event Hubs, Spark Structured Streaming, Event-Driven Pipeline Architecture
- **Cloud Platforms:** AWS (S3, Redshift, Glue, IAM, KMS, Lambda, EC2, RDS, Step Functions, Bedrock, SageMaker, OpenSearch), Microsoft Azure (ADLS Gen2, Synapse, Data Factory, Databricks, Event Hubs, HDInsight), Hybrid AWS/Azure Architectures
- **AI / ML & NLP Engineering:** Feature Store Design, ML Data Pipelines, Retrieval-Augmented Generation (RAG), AWS Bedrock, SageMaker, Transformer Fine-Tuning (Hugging Face, PyTorch, TensorFlow), Embedding Models, Vector Search (OpenSearch), LLM Evaluation & Guardrails, Clinical NLP (ICD-10, LOINC)
- **Orchestration & Workflow Management:** Apache Airflow, AWS Step Functions, Azure Data Factory Triggers, BMC Control-M
- **Data Governance & Security:** HIPAA & GDPR Compliance, PHI/PII Masking, Column-Level Security, RBAC, Okta Integration, Collibra, Alation, Data Lineage & Audit Controls
- **Data Quality & Observability:** Great Expectations, Custom Validation Frameworks, Freshness Monitoring, Anomaly Detection, Row-Count Reconciliation, Query History Audits, Pipeline Health Monitoring
- **DevOps & Infrastructure as Code:** Terraform, GitHub Actions, GitLab CI, Azure DevOps, dbt CI/CD, SQL linting (sqlfluff), Python testing (pytest), Cross-Account AWS Networking
- **Domain Expertise:** Healthcare (FHIR, HL7 v2, EHR, Imaging), Banking & FinTech (Fraud Detection, Risk Modeling, Regulatory Reporting), Customer 360, Executive KPI & Board-Level Analytics

EXPERIENCE

Senior Data Engineer – AI/ML | GE Healthcare *05/2023 – Present*

- Led the build of a cloud-native clinical data platform during GE HealthCare's spin-off transition, replacing fragmented on-prem Postgres warehouses with an AWS-based lakehouse (S3, Snowflake, Redshift) that became the unified foundation for analytics, AI, and clinical applications across business units.
- Re-architected legacy ETL workflows into scalable ELT pipelines using **PySpark on AWS Glue**, redesigning schemas into star models and incremental loading strategies that supported a 3x growth in data access and 4x increase in concurrent query volume without performance degradation.
- Engineered high-throughput ingestion pipelines for structured EHR extracts, HL7 v2 messages, device telemetry, and unstructured clinical notes using **Kafka**, **AWS Kinesis**, **and custom Python parsers (hl7apy, FHIR resources)**, processing millions of patient-level events daily.
- Designed and implemented a harmonized **FHIR-based clinical data layer** powering the **Edison Digital Health Platform**, transforming disparate EMR, imaging, and device feeds into standardized patient-centric resources consumable by AI and product teams.
- Introduced **dbt** into the Snowflake analytics layer, refactoring complex Spark SQL transformations into modular, version-controlled dbt models with built-in schema tests, source freshness monitoring, and incremental materializations, reducing runtime by 40% and enabling analytics teams to self-serve governed datasets.
- Built event-driven orchestration across ingestion, validation, transformation, and AI feature generation using **Airflow and AWS Step Functions**, coordinating dozens of interdependent pipelines with retry logic, SLA monitoring, and automated backfills.
- Developed a production-grade Retrieval-Augmented Generation (RAG) architecture using **AWS Bedrock, SageMaker-hosted embedding models (PyTorch), and Amazon OpenSearch vector indices**, allowing clinicians to query longitudinal patient histories across millions of radiology reports with grounded, source-attributed responses.
- Created secure prompt orchestration services in Python that injected structured FHIR patient context into LLM prompts, applied rule-based guardrails, and filtered unsafe generations, ensuring outputs aligned with clinical safety and compliance expectations.
- Fine-tuned transformer-based NLP models using **Hugging Face and TensorFlow/PyTorch** to extract diagnoses, medications, and procedures from unstructured notes, normalizing entities to ICD-10 and LOINC codes for downstream risk modeling and analytics.
- Implemented LLM evaluation pipelines combining automated factual-grounding checks with physician-reviewed benchmark datasets, measuring citation accuracy and response reliability before promoting model versions to production.
- Built near-real-time Spark streaming jobs to replace nightly batch refreshes, reducing clinical dashboard latency from 24 hours to under 10 minutes for time-sensitive use cases such as stroke imaging review and ICU monitoring.
- Established end-to-end data lineage and metadata visibility by integrating Snowflake and pipeline metadata into **Collibra and Alation**, exposing column-level traceability from raw ingestion through AI model feature inputs for audit and compliance teams.
- Implemented automated data quality enforcement using **Great Expectations and custom rule engines**, validating completeness, referential integrity, FHIR conformance, and medical coding accuracy, with anomaly reporting routed directly to engineering and product stakeholders.
- Designed PHI-aware security controls including fine-grained IAM policies, Snowflake RBAC, dynamic masking, encryption with AWS KMS, and tokenization layers for external model inference, maintaining HIPAA and GDPR compliance while enabling AI experimentation.

- Acted as technical lead during infrastructure separation from legacy GE systems, rebuilding CI/CD pipelines (GitHub Actions), isolating cross-account AWS networking, and eliminating shared dependencies to ensure the new standalone MedTech organization operated on fully independent data infrastructure.

Senior Data Engineer | SoFi

08/2020 – 04/2023

- Joined when SoFi was still operating like a high-growth lender and helped build the data foundation that supported its transition to a nationally chartered bank under SoFi, working across lending, banking, and investment products during the charter approval period.
- Designed and rebuilt core ELT pipelines in **Python (pandas, boto3, requests, PySpark)** and **SQL**, pulling raw event data from AWS S3, Aurora, and Kafka topics into Snowflake, then modeling it into finance-grade reporting tables used by Risk, Finance, and Compliance. I owned the pipelines from ingestion to BI layer, including failure handling and backfills.
- Served as the Snowflake lead for our domain, redesigning warehouse schemas using a hybrid **star schema + data vault** approach to support both regulatory reporting and product analytics; reduced average query time on risk dashboards from minutes to seconds by rewriting joins, clustering large fact tables, and tuning virtual warehouse sizing.
- Introduced **dbt** as the transformation layer for our credit and deposits data marts; built modular models with source freshness checks, custom macros, and automated documentation so analysts could trace metrics like "Net Charge-Off Rate" back to raw ledger events without Slack threads or tribal knowledge.
- Built and maintained **Apache Airflow** DAGs to orchestrate 200+ daily workflows; implemented retry logic, SLA alerts, and dependency graphs so downstream teams could see exactly why a pipeline failed and how long recovery would take. I also wrote custom Airflow operators to standardize Snowflake and S3 interactions.
- Implemented near real-time ingestion using **Apache Kafka** and **Kafka Connect**, streaming card transactions and fraud signals into Snowflake and Databricks; partnered with Risk Engineering so their fraud models could score transactions within seconds instead of relying on batch updates.
- Worked closely with ML engineers building fraud and credit risk models; created curated feature tables in **Databricks (PySpark, Delta Lake)** and versioned them for model reproducibility. Built validation scripts that compared model input distributions before and after major schema changes to prevent silent drift.
- Led data governance efforts required for operating as a bank: enforced column-level access controls in Snowflake, masked PII using dynamic data masking policies, and worked with Security to implement role-based access tied to Okta groups; prepared datasets used in audits tied to CCPA and bank regulatory reporting.
- Built CI/CD workflows in **GitLab CI** that ran dbt tests, SQL linting (sqlfluff), and Python unit tests (pytest) before merge; deployments were automated through environment-specific Terraform modules, reducing manual production changes and late-night hotfixes.
- Used **Terraform** to provision Snowflake roles, warehouses, S3 buckets, IAM policies, and Airflow infrastructure on **AWS (EC2, S3, RDS, Lambda, IAM)**; standardized infrastructure patterns so new product teams could onboard in days instead of weeks.
- Supported launch analytics for new products including credit cards and crypto trading; built reconciliation pipelines that tied transactional data to general ledger entries, giving Finance confidence that customer balances matched warehouse reporting.
- Partnered with BI teams using **Tableau** and **Power BI**; created certified data sources with documented metric definitions so executives could review board-level KPIs without asking whether the numbers were "finance-approved" or "product-approved."

- Mentored 4 junior and mid-level data engineers through code reviews and architecture sessions; pushed for smaller, testable pull requests and required data tests for every new model, which reduced production incidents tied to schema drift.
- Consolidated data feeds from the Galileo platform after its acquisition, mapping external event schemas into SoFi's internal models; wrote translation layers that normalized transaction types and customer identifiers, which allowed cross-product analytics for the first time.
- Established monitoring with custom Python health checks and Snowflake query history audits; built Slack alerts for abnormal row counts and freshness gaps, catching upstream API failures before they impacted regulatory reports.
- Acted as the bridge between engineering and non-technical stakeholders during the bank charter transition; translated regulatory reporting needs into concrete tables, lineage diagrams, and SLA commitments so leadership understood what data was reliable for filings and what was still in flight.

Data Engineer | Deloitte

04/2017 – 07/2020

- Led end-to-end design and build of cloud-first data platforms on **Microsoft Azure**, migrating clients from on-prem Oracle and SQL Server environments into Azure Data Lake Storage Gen2 with curated layers for reporting and data science.
- Built ingestion pipelines in **Python (pandas, pyodbc, requests)** and orchestrated them with **Azure Data Factory**, replacing legacy SSIS jobs and cutting manual refresh cycles from days to hours.
- Designed dimensional models (star and selective snowflake schemas) in **Azure Synapse Analytics** and **Snowflake**, working directly with finance and marketing stakeholders to translate reporting requirements into fact tables, conformed dimensions, and slowly changing dimensions (Type 2) using SQL window functions and incremental load patterns.
- Built distributed data processing jobs in **Azure Databricks (PySpark)** to handle multi-terabyte CRM and web interaction datasets; tuned Spark partitions, caching strategy, and join methods to reduce runtime by over 40% while staying within client SLA windows.
- Engineered ELT pipelines following a bronze–silver–gold layering approach before the term "medallion" became common; raw data landed in ADLS, transformation logic executed in Synapse or Databricks, and curated marts fed **Power BI** dashboards used by executive leadership.
- Partnered with data scientists to productionize machine learning datasets, creating feature tables for churn and fraud models; automated feature refresh using ADF triggers and parameterized pipelines, and versioned transformation logic in **Git** with pull request reviews and CI builds through **Azure DevOps**.
- Implemented data quality controls using Python validation scripts (Great Expectations-style custom checks), row count reconciliations, and referential integrity tests; surfaced failures to business owners through email alerts and dashboard indicators so issues were visible before board-level reporting cycles.
- Supported near real-time ingestion patterns using **Apache Kafka** (via Confluent connectors) for supply chain telemetry data; streamed events into Azure Event Hubs and processed them in Databricks Structured Streaming to power operational dashboards with minute-level latency.
- Contributed to hybrid cloud projects where certain workloads remained in AWS; integrated **Amazon S3** staging buckets with Azure pipelines and helped refactor Redshift SQL transformations into Synapse-compatible patterns, giving clients flexibility during phased cloud transitions.
- Wrote detailed technical design documents and data lineage diagrams in Visio and Lucidchart, explaining architecture decisions in plain language so finance directors and operations managers could follow how their numbers were produced and where controls were applied.

- Mentored junior engineers on SQL performance tuning, indexing strategies, and code review practices; introduced a reusable Python utilities package for logging, retry logic, and schema enforcement that became part of the team's standard project template.
- Delivered a Customer 360 program for a retail client by integrating Salesforce extracts, web clickstream data, and ERP transactions; built identity resolution logic using hashed keys and deterministic matching rules, enabling marketing teams to launch targeted campaigns with measurable lift in conversion rates.
- Worked in agile delivery teams (Scrum and SAFe environments), leading sprint planning for data workstreams and presenting live demos of pipeline builds and dashboard outputs to client stakeholders, which strengthened trust and shortened feedback cycles.

Junior Data Engineer | BMC Software

09/2015 – 02/2017

- Built and maintained batch and near-real-time pipelines pulling IT service data from Remedy, Oracle, PostgreSQL, and logs into Azure SQL Data Warehouse; used Python (pandas, requests, cx_Oracle) and complex SQL to standardize incident, change, and asset records for a single trusted view of operational data.
- Migrated reporting workflows from on-prem Oracle to Azure, creating Data Factory pipelines and Blob Storage staging; redesigned transformations from row-by-row PL/SQL to set-based SQL and Spark jobs on HDInsight, cutting nightly processing from hours to under one hour and reducing load failures.
- Automated job scheduling and dependency management with BMC Control-M, converting manual runbooks into version-controlled pipelines; documented job chains and error-handling rules, reducing production incidents and on-call interruptions.
- Supported early cloud adoption by exporting curated datasets from Azure to AWS S3 using Python/boto3, setting up secure transfers, IAM roles, and encryption policies to satisfy security and compliance requirements.
- Processed large volumes of logs with Hadoop MapReduce and Apache Spark; wrote Python and Bash parsers to clean data and created partitioned Hive tables; implemented validation checks—row counts, hash totals, null thresholds, referential integrity—with results logged for audit and upstream issue tracking.
- Produced technical documentation, ER diagrams, pipeline flow charts, and recovery steps; collaborated with analysts and early data science teams on anomaly detection and feature tables; worked in Linux environments writing Bash scripts, managing cron jobs, and debugging infrastructure issues to optimize cost and performance.

EDUCATION

Bachelor of Science – Southern Methodist University (2015)